

Saudi E-commerce Store Performance & Digital Presence Analysis

1- Project Objective:

Analyze data of 5,401 Saudi e-commerce stores to discover patterns of activity, ratings, digital presence, and data completeness, and transform raw data into actionable insights and recommendations that help understand store performance and support decision-making

2- Analytical Questions:

This project aims to answer a set of analytical questions that help understand the performance of Saudi e-commerce stores, their activity level, customer ratings, and digital presence, and transform the available data into clear insights that support business decision-making.

And through the analysis, the focus will be on the following questions:

- What is the status of active stores?
- Do ratings differ between active and inactive stores?
- What are the five most common business categories?
- Which categories recorded the highest percentage of active stores?
- Which categories recorded the highest average rating?
- What is the overall average rating for stores?
- Is there a relationship between the number of ratings and the average rating?
- What are the top 10 highest-rated stores, provided there are more than 10 ratings?
- To what extent are digital presence channels available to stores?
- What are the most common digital presence combinations?
- Does digital presence diversity relate to average rating?
- What are the most active business categories, and which categories deserve further study?
- Does digital presence diversity relate to store activity status?

3- Tools & Technologies Used:

Power BI - Power Query – Dax – Microsoft Excel – python (Pandas)

4- Data Overview:

The Saudi Ecommerce Data Set from Kaggle platform was used, containing information about 5,401 Saudi e-commerce stores distributed across 15 business categories and 13 columns. The data includes the store name in Arabic and English, store activity status, business category, ratings, and number of ratings, in addition to digital presence information such as website, Instagram, Twitter, email, and phone number.

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5- Data Understanding & Initial Exploration:

The first phase of analysis focuses on understanding the dataset and evaluating its initial structure and quality. This phase aims to identify data characteristics, detect potential issues in data quality, and ensure the suitability of available fields to answer the project's questions.

The initial exploration includes reviewing the data structure, data types, missing values, duplicate records, inconsistent values, and the distribution of key variables. The results of this phase will help guide data cleaning operations and analysis in the following stages.

6- Data Inspection:

After inspecting the dataset consisting of 5,401 stores, the data structure and its quality were evaluated, checking for missing values, duplicates, and inconsistent values, in addition to examining the key variables used in the analysis.

The inspection showed the following results:

- The dataset contains 5,401 records, in addition to 12 actual columns. An additional column (Unnamed: 12) was also discovered that is completely empty and will be deleted during the data cleaning phase.
- There are missing values in some fields, most importantly the store name in English with 7.24% and the commercial record CR with 64.64% and Instagram with 70.14% and Twitter with 86.32%. These values will be handled according to their importance during the cleaning phase.
- A fully duplicate record for the same store was discovered, where the store "Dorat Al-Matajer" appeared twice with the same data, and only one record will be kept during cleaning.
- The number of active stores reached 4,893 stores (90.59%), versus 508 inactive stores (9.41%).
- The data contains 15 commercial classifications. The "Other" classification is the most widespread, including 1,865 stores with a rate of (34.53%). This indicates that a large percentage of stores are classified under a general category, which may affect the accuracy of comparison between business activities.
- The inspection showed that 4,662 stores with a rate of (86.32%) have a Rating = 0, which is the same percentage of stores that have Num_Ratings = 0. This indicates that zero in the rating is associated with the absence of ratings, and not necessarily a negative rating for the store.

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- No Rating values outside the 0-10 range were detected, and no negative rating counts were found.
- All phone numbers in the data consist of 9 digits, and no clear issues were detected in the format of email addresses during the inspection.

The inspection showed that the dataset is generally suitable for analysis, with some data quality issues that must be addressed in the cleaning phase, most notably the empty column, the duplicate record, and the missing values. After addressing these issues, the data will be more ready for analysis and extracting the results and indicators required for the project.

7- Data Cleaning:

After completing the data inspection, a set of cleaning operations was implemented to improve data quality and prepare it for analysis, while preserving the original information and not deleting any data unless there is a clear reason.

The cleaning process included the following:

- Deleting the column "Unnamed: 12" because it was completely empty and does not contain any useful information.
- Removing the fully duplicated record belonging to one of the stores.
- Using Trim to remove extra spaces from text fields.
- Adjusting data types to suit the nature of each column; where "CR" and "Phone Number" were converted to Text, and "Rating" to Decimal Number, and "Num_Ratings" to Whole Number.
- Keeping the missing values in fields such as "CR" and "Instagram" and "Twitter" instead of deleting rows, because they represent unavailability of information and can be utilized in the analysis.
- Verifying the validity of "Rating" values and ensuring there are no values less than 0 or greater than 10.
- Checking "Num_Ratings" and ensuring there are no negative values.
- Examining email addresses, phone numbers, and links to ensure there are no obvious problems in their formatting.

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After implementing these operations, the data became more organized and ready to move to the data analysis phase and extract the main results and indicators.

8- Data Analysis & Results:

After completing the data cleaning and preparation phase, the transition was made to the data analysis phase with the aim of extracting the main patterns and indicators related to store activity, distribution of business categories, customer ratings, and the digital presence of stores.

The analysis was implemented using Power BI and DAX, along with counting operations, calculating percentages, averages, filtering, sorting, and comparison between groups.

The analysis was built around 13 main analytical questions, distributed across four axes:
Store activity analysis - Business categories analysis - Ratings analysis - Digital presence analysis

Question 1: What is the status of store activity?

Finding:

Active stores constitute the vast majority of the sample, with their number reaching 4,892 stores at a rate of 90.59%, while the number of inactive stores reached 508 stores at a rate of 9.41%.

Question 2: Do ratings differ between active and inactive stores?

Finding:

The results showed that active stores have a higher average rating than inactive stores. After excluding stores that have no actual ratings, the average rating of active stores reached 9.01/10, compared to 7.47/10 for inactive stores, with a difference of 1.54 points. This is considered a descriptive finding from the data, and does not necessarily prove that activity is the cause of the higher rating.

Question 3: What are the five most common business categories?

Finding:

The most common business categories were Other with 1,864 stores, followed by E-marketing with 1,485 stores, then Women's Supplies with 522 stores, and Electronics and Accessories with 394 stores, and finally E-solutions with 268 stores.

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Question 4: What categories recorded the highest percentage of active stores?

Finding:

The results showed that Academic Services recorded the highest percentage of active stores at 98.59%, followed by Coiffure and Beauty at 96.97%, then Crafts and Handicrafts at 95.71%, while both Cars and Event Planning recorded 95.65%.

Question 5: What categories recorded the highest average rating?

Finding:

The results showed that the highest average rating reached 10.00/10, with a tie between several categories. Most notably Cars, Real Estate, Event Planning, Design and Printing, and Photography. This means that these categories recorded the highest average rating among categories that have actual ratings.

Question 6: What is the overall average rating for stores?

Finding:

The average rating of stores that have actual ratings reached 8.60/10, which indicates a relatively high overall rating level among stores that have recorded ratings.

Question 7: Is there a relationship between the number of ratings and the average rating?

Finding:

The data shows a weak inverse relationship between the number of ratings and the average rating. When there are few ratings, such as 0-20 ratings, ratings range approximately between 4 and 10, with many concentrated at 9 and 10.

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With an increase in the number of ratings, a general tendency for the rating to decrease appears; at around 50 ratings, ratings such as 8.4, 7.8, and 8.9 appeared, and at around 100 ratings, ratings such as 8.8 and 5.7 appeared, while a point appeared at around 170 ratings with a rating close to 5.8.

This indicates that an increase in the number of ratings tends to be associated with a decrease in the rating, but the relationship is weak, and the number of ratings cannot be considered a direct cause of the decrease in rating.

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Question 8: What are the top 10 highest-rated stores, provided there are more than 10 ratings?

Finding:

After applying the condition of having more than 10 ratings, the top 10 stores came as follows:

- Asrar Al-Enaya 10.00 / 10 - 11 ratings
- Perfect Story 10.00 / 10 - 15 ratings
- Selling premium usernames for social media programs 10.00 / 10 - 11 ratings
- Salat Al-Farah 10.00 / 10 - 29 ratings
- Sahilha Laltaqeeb 10.00 / 10 - 11 ratings
- Fire for monthly subscriptions Netflix and Shahid 10.00 / 10 - 94 ratings
- Ibrahim Shop Store 10.00 / 10 - 16 ratings
- Aromask Store 10.00 / 10 - 14 ratings
- Al-Deem Store for natural skin and hair products 10.00 / 10 - 14 ratings
- Riyadh Store 10.00 / 10 - 11 ratings

The results show that all ten stores received a full rating of 10.00/10, while the number of ratings differed clearly, ranging between 11 and 94 ratings. This reflects a tie in the highest ratings with a difference in the volume of ratings on which these results are based.

Question 9: To what extent are digital presence channels available to stores?

Finding:

The results showed that 100% of stores have a Website, Email, and Phone Number, which indicates that these basic channels are available to all stores in the sample. In contrast, the percentage of stores that have Instagram reached about 30%, while the percentage of stores that have Twitter reached about 14%.

The results indicate that the basic digital presence is widely available in the data, while the use of social media platforms varies, with Instagram being more widespread than Twitter.

Question 10: What are the most common digital presence combinations?

Finding:

The results showed that 66.44% of stores have only a Website without Instagram or Twitter, while 19.87% of stores have a Website and Instagram without Twitter.

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Also, 3.69% have a Website and Twitter without Instagram, while 10.00% of stores have all three channels together.

The results indicate that the majority of stores rely on a Website as the primary digital presence channel, while the percentage of stores that combine the website and multiple social media platforms is lower.

Question 11: Does digital presence diversity relate to average rating?

Finding:

The results showed that stores with more diverse digital presence recorded a higher average rating. The average rating of stores that have only a Website reached 8.15, compared to 9.59 for stores that have a Website and Instagram, and 9.23 for stores that have a Website and Twitter, while stores that have all three channels recorded the highest average of 9.71/10.

This indicates that stores with more presence across digital channels recorded higher ratings for the store itself in this sample. This does not mean that customers rated the digital channels, but that the store's rating was higher among stores that have more diverse digital presence.

Question 12: What are the most active business categories, and which categories deserve further study?

Finding:

The results showed that the level of store activity is generally high, with the percentage of active stores reaching 90.59% of total stores, versus 9.41% of inactive stores.

And when comparing business categories by activity rate, Academic Services came in first place with an activity rate of 98.59%, followed by Coiffure and Beauty with 96.97%, then Crafts and Handicrafts with 95.71%.

As for inactive stores, the E-marketing category recorded the highest percentage at 18.59%, followed by Photography at 14.29%, then Real Estate at 13.64%.

Therefore, these categories can be considered areas that deserve further study to understand the reasons for the high rate of inactivity in them, and whether there are opportunities to improve the continuity and performance of stores.

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In general, the data indicates that most categories have a high level of activity, with some categories that need more attention and analysis. And a decrease in activity alone cannot be considered evidence of a confirmed business opportunity, because accurately determining growth opportunities requires additional data such as sales, demand, number of customers, and market size.

Question 13: Does digital presence diversity relate to store activity status?

Finding:

The results showed that stores that have more diverse digital presence recorded a higher activity rate. The percentage of active stores that have Website + Instagram + Twitter reached about 97%, compared to 87.7% for stores that have only Website.

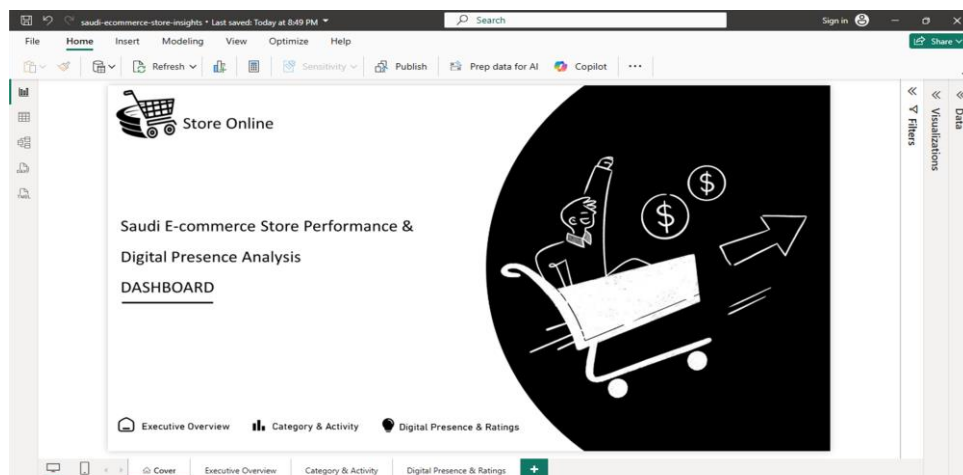
These results indicate the existence of a descriptive correlation between the diversity of digital presence and the store's activity status; where stores that have more than one digital channel were more active in this sample. However, digital presence cannot be considered a direct cause of increased store activity, but rather an indicator associated with activity status according to the available data.

9- Interactive Dashboard (Power BI Dashboard):

An interactive dashboard was developed using Power BI to display the analysis results in a clear and organized manner, providing indicators and charts that help explore store activity, business categories, ratings, and digital presence.

Cover Page:

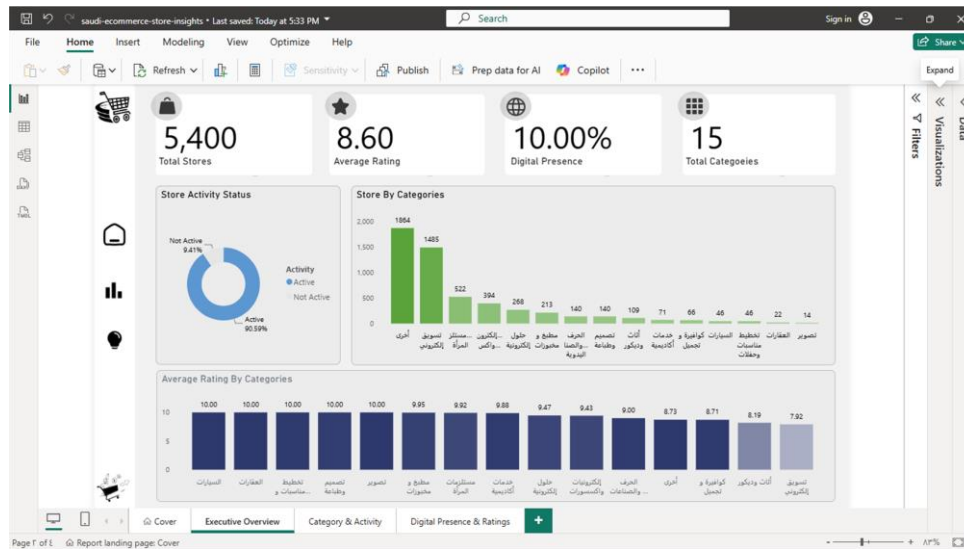
A professional cover page was designed that reflects the subject of the project and provides a clear entry point to the dashboard.



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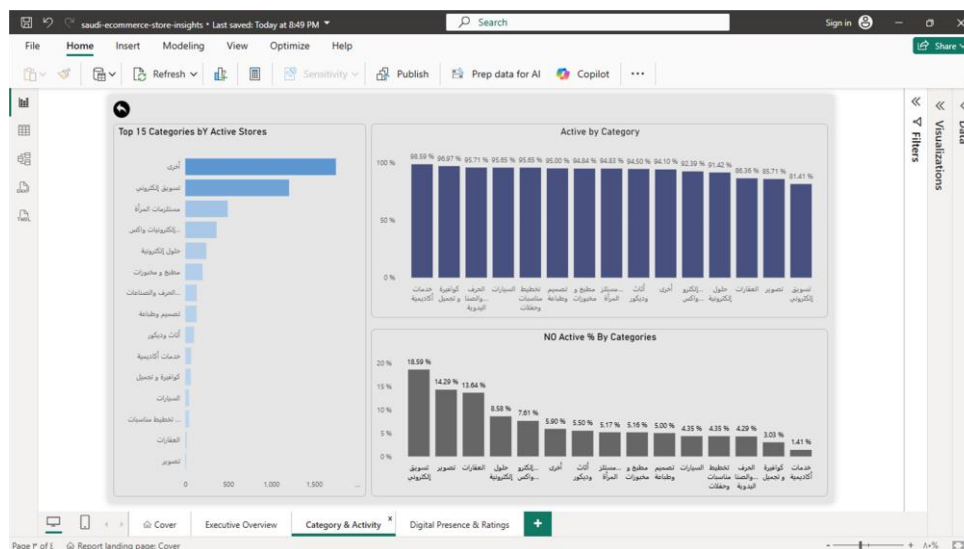
Executive Overview:

The page displays the most important indicators and main results, including total stores, active stores, activity rate, average rating, and number of business categories, in addition to the most important analytical charts.



Category & Activity:

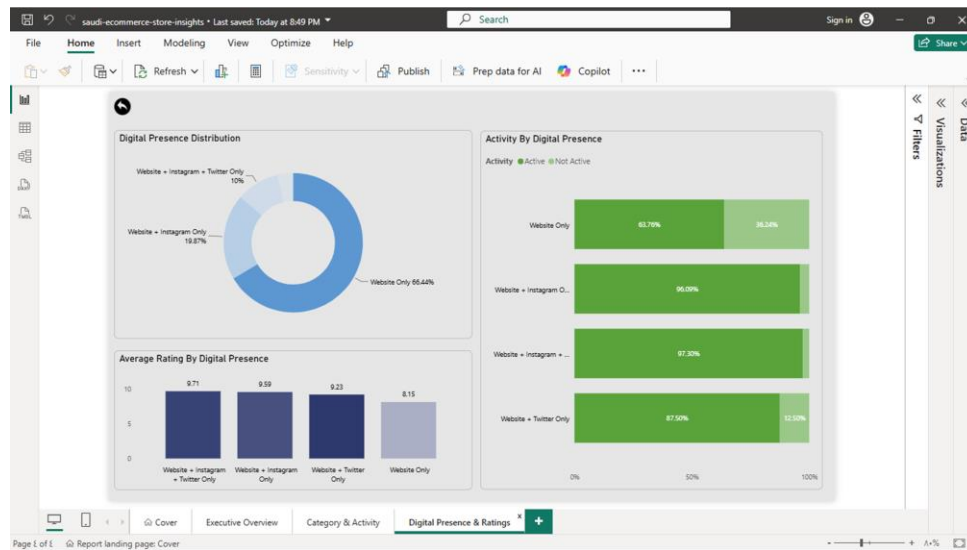
The page focuses on comparing business categories by activity level, displaying activity and inactivity percentages and the number of stores in each category.



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Digital Presence & Ratings:

The page displays the patterns of digital presence for stores and its relationship to rating and activity levels, using a set of interactive charts.



10- Key Findings:

Analysis of data from 5,401 Saudi e-commerce stores showed the following results:

- The percentage of active stores reached 90.59% versus 9.41% for inactive stores.
- The average rating of stores reached 8.60 out of 10.
- The analysis included 15 business categories, with different levels of activity and rating between categories.
- The Academic Services category recorded the highest activity rate, reaching 98.59%.
- The E-marketing category recorded the highest percentage of inactive stores, reaching 18.59%.
- The results showed a difference in average rating by business category, where some categories recorded high averages compared to others.
- Stores that have Website and Instagram and Twitter recorded an average rating of 9.71 out of 10, compared to 8.15 out of 10 for stores that have only Website.
- The activity rate in stores with diverse digital presence reached about 97%, compared to about 87.7% for stores that have only Website.
- The results indicate the existence of a descriptive correlation between the diversity of digital presence and the level of activity and ratings, without proving that digital presence is the direct cause of the increase in these indicators.

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11- Recommendations:

Based on the analysis results, the following recommendations can be provided:

- Study the reasons for the high percentage of inactive stores in some categories, especially E-marketing, Photography, and Real Estate.
- Continuously monitor the activity status of stores by business category.
- Encourage stores to diversify their digital presence across websites and social media platforms.
- Monitor customer ratings alongside store activity indicators.
- Improve data quality and update it periodically to ensure analysis accuracy.
- Add sales, revenue, orders, and customer data in the future to obtain a deeper analysis of store performance.
- Develop the dashboard to support periodic monitoring and comparison of changes in store activity and ratings.

12- Project Limitations:

The project faced a number of limitations that must be taken into consideration when interpreting the results, most notably:

- Lack of sales, revenue, orders, and customer data.
- Lack of detailed geographic information about store locations.
- Existence of missing values in some fields, especially digital presence data and the commercial record.
- Existence of rating values equal to zero, which may reflect the absence of actual ratings.
- The data represents a specific time period, and does not show changes in store performance over time.
- The results show relationships and patterns within the data, and do not prove the existence of a causal relationship between digital presence and store activity or its rating

Accordingly, the project results focus on descriptive and exploratory analysis of the available data, and do not represent a comprehensive evaluation of the commercial performance of the stores.

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13- Conclusion & Future Development:

Conclusion

This project presented a comprehensive analysis of data from 5,401 Saudi e-commerce stores, with the aim of understanding patterns of store activity, their distribution by business categories, rating levels, and the extent of their digital presence diversity. The project went through integrated phases that began with examining the data, cleaning and preparing it, then conducting the analysis and extracting indicators, culminating in the design of an interactive dashboard using Power BI, Power Query, and DAX.

The results showed a high level of store activity in general, along with clear differences between business categories in activity and rating levels. The analysis also showed the existence of a descriptive correlation between the diversity of digital presence and both store activity and average rating. These results provide a clearer understanding of the patterns existing in the data, taking into account the limits of the available data and not interpreting relationships as direct causes.

Future Development

The project can be expanded in the future by integrating additional data that provides a more comprehensive picture of store performance, such as:

- Sales, revenue, and orders data.
- Customer data and their purchasing behavior.
- Geographic data for stores.
- Visit and interaction data with websites and social media platforms.
- Temporal data to track the evolution of store performance.
- Adding advanced analytics and predictive models to support decision-making.

Through these additions, the project can be developed from a descriptive analysis of data into a more comprehensive analytical system for studying the performance of e-commerce stores and their future trends.